Experiment No. 19 – Erosion Morphological Operation Using OpenCV

1. Aim

To implement the Erosion Morphological Operation technique using OpenCV in Python.

2. Program

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

# Read image
filename = list(uploaded.keys())[0]
image = cv2.imdecode(
    np.frombuffer(uploaded[filename], np.uint8),
    cv2.IMREAD_COLOR
)

if image is None:
    print("Error: Image not loaded")
else:
    # Convert image to grayscale
    gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
# Convert image to binary
_, binary = cv2.threshold(
    gray, 127, 255, cv2.THRESH_BINARY
)

# Create structuring element
kernel = np.ones((5, 5), np.uint8)

# Apply erosion operation
erosion = cv2.erode(
    binary, kernel, iterations=1
)

# Display images
plt.figure(figsize=(12, 6))

plt.subplot(1, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 3, 2)
plt.imshow(binary, cmap="gray")
plt.title("Binary Image")
plt.axis("off")
```

```
plt.subplot(1, 3, 3)
plt.imshow(erosion, cmap="gray")
plt.title("Erosion Image")
plt.axis("off")

plt.tight_layout()
plt.show()
```

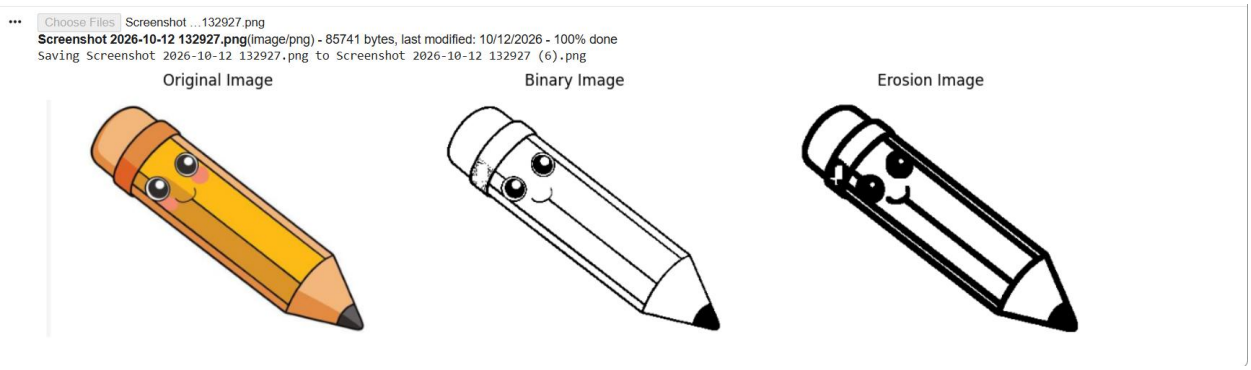
3. Expected Output

The output displays:

- Original Image.
- Binary Image.
- Erosion Image.

In the eroded image, the white foreground objects become smaller because erosion removes pixels from their boundaries.

4. Output



The erosion operation is successfully applied to the binary image. The foreground objects are reduced in size.

5. Result

Thus, the Erosion Morphological Operation was successfully implemented using Python and OpenCV.